

Module 14: Debugging and Problem Solving with JavaScript

Practice Tasks (Find the Bug, Fix It)

Instructions:

- Run the code first and see what actually happens.
- Write a one-line comment explaining what the bug is, before fixing it.
- Fix the bug so the output matches the "Expected Output."
- Every function must still **return** its answer (not just console.log it).

Task 14-1: Spot the Error Type

Below are three small broken snippets. For each, identify the error type (Syntax / Runtime / Logic), then fix it.

Snippet A

```
function sayHi(name) {  
  console.log("Hi, " + name // find the bug  
}  
  
sayHi("Mira");
```

Input	Expected Output
sayHi("Mira")	Hi, Mira

Snippet B

```
let word = "hello";  
console.log(word.toUpperCase()); // find the bug
```

Input	Expected Output
word.toUpperCase()	HELLO

Snippet C

```
function square(num) {  
  return num + num; // find the bug  
}  
  
console.log(square(4)); // should be 16
```

Input	Expected Output
square(4)	16

Starter Snippet (fix all three below):

```
// Bug type: ____  
function sayHi(name) {  
  console.log("Hi, " + name)  
}  
sayHi("Mira");  
  
// Bug type: ____  
let word = "hello";  
console.log(word.toUpperCase());  
  
// Bug type: ____  
function square(num) {  
  return num + num;  
}  
console.log(square(4));
```

Task 14-2: Debug a Broken Conditional

Problem Statement:

This function should return "Pass" if a score is 50 or above, otherwise "Fail". It gives the wrong result for a score of exactly 50.

Input	Expected Output
50	"Pass"
40	"Fail"

Starter Snippet (buggy):

```
// Bug: find why score = 50 doesn't return "Pass"

function checkScore(score) {
  if (score > 50) {
    return "Pass";
  } else {
    return "Fail";
  }
}

console.log(checkScore(50)); // Expected: "Pass" (currently wrong)
console.log(checkScore(40)); // Expected: "Fail"
```

Task 14-3: Debug a Broken Loop

Problem Statement:

This function should return the count of numbers from 1 to n. Right now it never stops (infinite loop) — find out why.

Input	Expected Output
5	5

Starter Snippet (buggy):

```
// Bug: this loop never ends — find the missing piece

function countUpTo(n) {
  let count = 0;

  for (let i = 1; i <= n;) { // something is missing here
    count = count + 1;
  }

  return count;
}

console.log(countUpTo(5)); // Expected: 5 (currently freezes/never finishes)
```

Task 14-4: Debug a Broken Array Problem

Problem Statement:

This function should return the first element of an array. It currently returns the wrong item.

Input	Expected Output
[10, 20, 30]	10

Starter Snippet (buggy):

```
// Bug: find why this doesn't return the first element

function getFirstElement(arr) {
  return arr[1]; // find the bug
}

console.log(getFirstElement([10, 20, 30])); // Expected: 10 (currently returns 20)
```

Task 14-5: Debug a Broken Function

Problem Statement:

This function should return the difference between two numbers (a - b). It currently swaps the parameters somewhere and gives a wrong result.

Input	Expected Output
10, 3	7

Starter Snippet (buggy):

```
// Bug: find why the result comes out negative

function subtract(a, b) {
  return b - a; // find the bug
}

console.log(subtract(10, 3)); // Expected: 7 (currently returns -7)
```

Task 14-6: Rubber Duck Debugging

Problem Statement:

Explain this function out loud, line by line, to find the bug. It should return true if a string is empty, otherwise false.

Input	Expected Output
""	true
"hi"	false

Starter Snippet (buggy):

```
// Bug: explain this line out loud – what does str.length actually equal for ""?  
  
function isEmpty(str) {  
  if (str.length = 0) { // find the bug  
    return true;  
  } else {  
    return false;  
  }  
}  
  
console.log(isEmpty("")); // Expected: true  
console.log(isEmpty("hi")); // Expected: false
```

Task 14-7: Debug a Multi-Part Bug

Problem Statement:

This function should return an array of only the odd numbers from the input array. It currently has two separate bugs — fix them one at a time.

Input	Expected Output
[1, 2, 3, 4, 5, 6]	[1, 3, 5]

Starter Snippet (buggy):

```
// This code has TWO bugs. Fix one, re-run, then find the next.  
  
function getOddNumbers(numbers) {
```

```

let odds = [];

for (let i = 0; i <= numbers.length; i++) { // bug #1 is here
  if (numbers[i] % 2 = 1) { // bug #2 is here
    odds.push(numbers[i]);
  }
}

return odds;
}

console.log(getOddNumbers([1, 2, 3, 4, 5, 6])); // Expected: [1, 3, 5]

```

Task 14-8: Apply the Full Debugging Process

Problem Statement:

This function should return how many vowels are in a string. Go through the full checklist: does it run? does it crash? is the answer wrong? isolate it with `console.log`, then fix it.

Input	Expected Output
"orange"	3

Starter Snippet (buggy):

```

// Apply the full debugging process to find this bug

function countVowels(str) {
  let vowels = "aeiou";
  let count = 0;

  for (let i = 0; i <= str.length; i++) { // find the bug
    if (vowels.includes(str[i])) {
      count = count + 1;
    }
  }

  return count;
}

console.log(countVowels("orange")); // Expected: 3

```

Task 14-9: Final Homework Set (Module Summary)

Each task below has one hidden bug. Use the full checklist from 14-10: run it, find the error type, isolate with console.log or breakpoints, fix one bug at a time.

14-9A: Temperature Converter

Problem Statement: Should convert Celsius to Fahrenheit using the formula $(C \times 9/5) + 32$.

Input	Expected Output
0	32

Buggy Code:

```
function celsiusToFahrenheit(celsius) {  
  return celsius * 9 / 5 + 3; // find the bug  
}  
  
console.log(celsiusToFahrenheit(0)); // Expected: 32
```

14-9B: Array Sum Skips a Number

Problem Statement: Should return the sum of all numbers in the array.

Input	Expected Output
[1, 2, 3, 4]	10

Buggy Code:

```
function sumArray(numbers) {  
  let total = 0;  
  
  for (let i = 1; i < numbers.length; i++) { // find the bug  
    total = total + numbers[i];  
  }  
  
  return total;  
}  
  
console.log(sumArray([1, 2, 3, 4])); // Expected: 10
```

14-9C: String Repeat

Problem Statement: Should return a string repeated a given number of times.

Input	Expected Output
"ab", 3	"ababab"

Buggy Code:

```
function repeatString(str, times) {
  let result = "";

  for (let i = 0; i < times; i++) {
    result = str; // find the bug
  }

  return result;
}

console.log(repeatString("ab", 3)); // Expected: "ababab"
```

14-9D: Object Property Update

Problem Statement: Should update a person's age property and return the updated object.

Input	Expected Output
{ name: "Lee", age: 20 }, 21	{ name: "Lee", age: 21 }

Buggy Code:

```
function updateAge(person, newAge) {
  person.age = person; // find the bug
  return person;
}

console.log(updateAge({ name: "Lee", age: 20 }, 21));
// Expected: { name: "Lee", age: 21 }
```